

● EASY

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

05

10 answers · Rationale-rich · Teach from doc

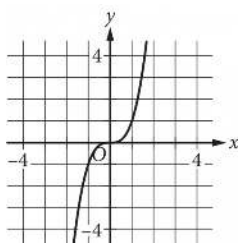
Q1**D****D. 3**

Choice D is correct. Because the graph of $y = f(x)$ is shown, the value of $f(0)$ is the value of y on the graph that corresponds with $x = 0$. When $x = 0$, the corresponding value of y is 3. Therefore, the value of $f(0)$ is 3.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q2**B****B.**

Choice B is correct. Each pair of values shown in the table gives the ordered pair of coordinates for a point that lies on the graph that represents the relationship between x and y in the xy -plane: $(0,0)$, $(1,1)$, $(2,8)$, and $(3,27)$. Only the graph in choice B passes through the points listed in the table that are visible in the given choices.

Choices A, C, and D are incorrect. None of these graphs passes through the point $(1,1)$.

Q3**B****B.** 0, 2

Choice B is correct. The y-intercept of a graph in the xy -plane is the point at which the graph crosses the y -axis. The graph shown crosses the y -axis at the point 0, 2. Therefore, the y-intercept of the graph shown is 0, 2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**A****A.** -152

Choice A is correct. The value of $f(0)$ is the value of $f(x)$ when $x = 0$. The function f is defined by $f(x) = 10x^2 - 32x - 152$. Substituting 0 for x in this equation yields $f(0) = 10(0)^2 - 32(0) - 152$. This equation can be rewritten as $f(0) = 100 - 0 - 152$, or $f(0) = -152$. Therefore, the value of $f(0)$ is -152.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**77**

The correct answer is 77. It's given that the function f is defined by $f(x) = x^2 + x + 71$.

Substituting 2 for x in function f yields $f(2) = 2^2 + 2 + 71$, which is equivalent to $f(2) = 4 + 2 + 71$, or $f(2) = 77$. Therefore, the value of $f(2)$ is 77.

Q6**D**

D. 22

Choice D is correct. The value of $g4$ is the value of gx when $x = 4$. Substituting 4 for x in the given equation yields $g4 = 4 + 18$, which is equivalent to $g4 = 22$, or $g4 = 22$. Therefore, the value of $g4$ is 22.

Choice A is incorrect. This would be the value of $g4$ if function g was defined by $gx = -18$, not $gx = x + 18$.

Choice B is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x$, not $gx = x + 18$.

Choice C is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x + 18$, not $gx = x + 18$.

Q7**3**

The correct answer is 3. A y -intercept of a graph in the xy -plane is a point x, y on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is 3. Therefore, the y -coordinate of the y -intercept of the graph shown is 3.

Q8**B****B.** \$ 13,632

Choice B is correct. For the graph shown, the horizontal axis represents the number of years after a certain sport utility vehicle is first purchased, and the vertical axis represents the predicted value, in dollars, of the sport utility vehicle. According to the graph, 3 years after the sport utility vehicle is purchased, the predicted value of the sport utility vehicle is between \$ 10,000 and \$ 15,000. Of the given choices, only \$ 13,632 is between \$ 10,000 and \$ 15,000. Therefore, \$ 13,632 is closest to the predicted value of the sport utility vehicle 3 years after it is first purchased.

Choice A is incorrect. This is closest to the predicted value of the sport utility vehicle 5 years after it is first purchased.

Choice C is incorrect. This is closest to the predicted value of the sport utility vehicle 1 year after it is first purchased.

Choice D is incorrect. This is closest to the predicted value of the sport utility vehicle when it is first purchased.

Q9**7**

The correct answer is 7. It's given that the x -intercept of the graph shown is $x, 0$. The graph passes through the point $7, 0$. Therefore, the value of x is 7.

Q10

B

B. $n = 53(2)^d$

Choice B is correct. It's given that each day for five days after a social media channel posted a video, the number of subscribers doubled from the number the previous day. Since the number of subscribers doubled each day, the relationship between n and d can be represented by an exponential equation of the form $n = ab^d$, where a is the number of subscribers at the time of posting the video and the number of subscribers to the channel increases by a factor of b each day. It's given that at the time of posting the video, the channel had 53 subscribers. Therefore, $a = 53$. It's also given that the number of subscribers doubled, or increased by a factor of 2, from the number the previous day. Therefore, $b = 2$. Substituting 53 for a and 2 for b in the equation $n = ab^d$ yields $n = 532^d$.

Choice A is incorrect. This equation gives the total number of subscribers to a channel for which the initial number of subscribers was 1 and the number increased each day by 53 times the number from the previous day.

Choice C is incorrect. This equation gives the total number of subscribers to a channel for which the number of subscribers each day was half the number from the previous day, rather than double the number.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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